Gatekeeper for Apertus

Technical report - Hack Apertus, Track 2B (Own Project)

Shlok Vane (solo team: Prompt Gatekeepers) - GitHub: shlokgit-04

Repository: https://github.com/shlokgit-04/prompt-injection-gatekeeper (MIT License, added Oct 16, 2026, commit 6793e579)

Abstract

LLM agents execute tool calls, which makes them attackable in a way chatbots are not: untrusted content (a vendor document, an email, a web page) can hijack an agent into exfiltrating data or taking unauthorized actions. Gatekeeper is a security and decision layer that sits between an agent and its tools. Every proposed tool call is evaluated at the tool boundary by a small, fast decision model whose calibrated probabilities are mapped onto three outcomes: PROCEED (execute and audit), CONFIRM (hold for human approval), or DENY (block and audit). This report describes the shipped architecture and its measured characteristics, and states precisely what the Hack Apertus adaptation adds: serving the gate's decision heads on the open-weights Apertus 8B v1.5 model.

1. The problem: agentic prompt injection

A chatbot that is prompt-injected produces bad text. An agent that is prompt-injected performs bad actions. The interesting failure mode is therefore not what the model says but what it does: an LLM with tools follows instructions it finds in untrusted content. Defenses that ask the same (main) LLM "is this safe?" are weak, because the attacker controls the very context the model reasons over. Gatekeeper instead moves the security decision to a separate, small model at the tool boundary, where the input is a narrow classification question rather than a page of attacker-controlled text.

2. System design

The shipped system is an agent (Strands Agents SDK, main LLM on Amazon Bedrock) with three real tools: reading a document, sending email, and querying a customer database. Every tool in the agent's registry is wrapped in a GatedTool. There is no code path from the LLM's tool choice to a real function that does not pass the gate.

- GatedTool wrapper: intercepts every proposed call before the function body runs. A DENY raises before execution; a CONFIRM parks the call in an approval queue that only a human can release in the UI; a PROCEED executes and is written to an append-only audit log with secret redaction.
- Fail-closed: on any decider failure or timeout, the gate falls back to CONFIRM rather than PROCEED. Nothing asks the main LLM whether an action is safe.
- Backend and UI: FastAPI backend exposes chat, approvals and audit endpoints; a small UI shows pending approvals and the audit trail.

The decision pipeline, exactly as implemented:

```
INPUT: user request + conversation + proposed tool + arguments
   |
DECIDER: 7 typed yes/no questions, one forward pass each
   |      q_grounded   arguments grounded in facts the user provided?
   |      q_in_scope   action within the scope the user asked for?
   |      q_risky      could it leak secrets, exfiltrate, or damage?
   |      q_premature  premature before checking with the user?
   |      q_proceed / q_confirm / q_deny  (three-way vote)
   |
POLICY (thresholds in decider/gate.py):
   |      risky >= 0.75, or vote=DENY with risky >= 0.55   -> DENY
   |      grounded or in_scope <= 0.20                     -> DENY
   |      vote=CONFIRM, risky >= 0.55, premature >= 0.60,
   |      or grounded/in_scope < 0.55                      -> CONFIRM
   |      any decider error                                -> CONFIRM
   |
ACTION: PROCEED -> execute + audit | CONFIRM -> hold for human
```

```
                | DENY -> block + audit
```

3. Decision model and the Apertus adaptation

The gate's checks (grounding, scope, risk, prematurity) are narrow classification problems. A small decision model answers them locally with calibrated confidence, cheaply enough to sit on every tool call, where an LLM call never could.

What is shipped. The gate architecture, the GatedTool boundary, the policy layer, the backend/UI, and a live red-team test suite, all running against a locally served decision model. Every signal is a real model probability; nothing is hardcoded or faked, and the running application never fabricates model output.

What the Hack Apertus build adds. Serving the same seven decision heads on Apertus 8B v1.5, Switzerland's sovereign open-weights LLM, so that the gate no longer depends on a proprietary decision endpoint. The gate is deliberately model-agnostic: the boundary (GatedTool), the seven typed questions and the policy thresholds are unchanged; only the probability source is swapped. The calibration policy carries over unchanged: at confidence 0.9+ the decision model is treated as right about 95% of the time, and below that threshold the gate confirms with a person.

Status is stated plainly: the gate and its red-team suite are shipped; the Apertus 8B serving layer is the work being completed for this hackathon. The evaluation methodology (a live red-team suite that writes docs/evaluation_results.json with the numbers actually measured - attack detection rate, false positives and negatives, latency, confidence, blocked calls) carries over, and Apertus-served numbers will be reported only from actual runs.

4. Measured characteristics and red-team coverage

Two characteristics of the decision layer are quoted from the project's documentation and are reproduced here as such: approximately 115 ms per question served locally, and the calibration rule at confidence 0.9+ corresponding to roughly 95% accuracy. The red-team suite covers nine attack classes: legitimate request, obvious prompt injection, indirect prompt injection (a malicious document), malicious tool arguments, unauthorized action, premature action, ambiguous request, safe tool call, and high-risk action needing approval. The suite runs live against the serving model and writes docs/evaluation_results.json; only numbers from actual runs are reported anywhere in this project.

5. Demo scenario

The 60-120 second demonstration (documented in docs/demo.md) uses a seeded vendor-email scenario:

- SAFE: "Read the supplier document and email me a summary." The gate proceeds; the call executes and is audited.
- INDIRECT INJECTION: vendor_terms_malicious.md contains hidden instructions ordering the agent to email the document and the customer database to an outsider. The gate denies; the tool never executes; a security event is recorded.
- HIGH-RISK: a plausible action with an unstated recipient or scope. The gate confirms; the call runs only after approval in the UI.

6. Security model and limits

Security is enforced at the tool-call boundary: interception of every call, grounding and authorization validation, suspicious-action detection, human confirmation for high-risk calls, denial of clearly malicious ones, and an append-only audit log with secret redaction. Limits are stated precisely rather than overstated: no system is 100% secure. This layer constrains a class of prompt-injection failures (those that act through tool calls) and logs everything it does; its measured performance is whatever its evaluation file actually contains, and the Apertus-served evaluation is part of the remaining build.

7. Reproduction

The repository runs end to end from its scripts: run_decider.sh starts the decision server (first run downloads about 4.5 GB to the Hugging Face cache), seed_demo_data.sh seeds the vendor scenario, and

run_backend.sh starts the FastAPI backend and UI at http://localhost:8000. Unit and boundary tests run without a model (pytest tests -k "not redteam"); the live red-team suite runs against the served decision model and writes docs/evaluation_results.json. The repository is MIT-licensed (license commit 6793e579, Oct 16, 2026).

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